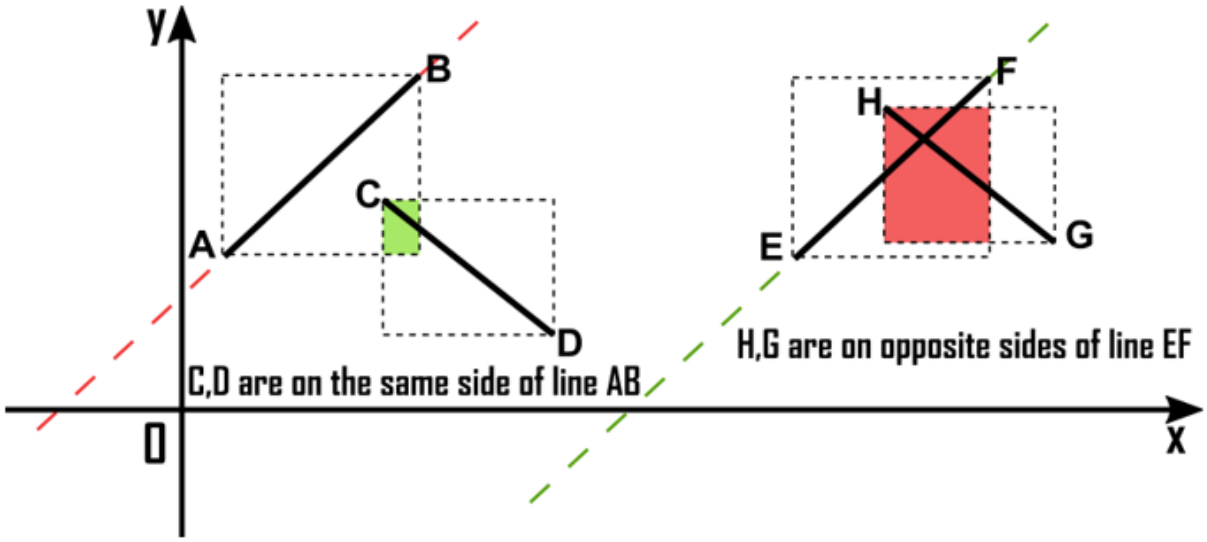


Figure 15.17 Segment intersection: if bounding boxes have a non-null intersection, the discriminant becomes whether or not the vertices of segment `CD` (or `HG`) are on opposite sides of the line passing through the segment `AB` (`EF`), and vice versa.

So, we have an asymmetric situation:

- If we find out that the projections of the segments do not intersect over one of the two axes, we can conclude that the segments do not intersect.
- Otherwise, we can't make any assumption, and we need to further investigate.

We can do that in many ways, for instance verifying that the extremes of a segment fall on the sides of the line passing through the other segment, as shown in figure 15.17.

This, however, requires a bit too much fiddling with edge cases (f.i. parallel segments), so I prefer a different, more elegant, method I recently discovered.